INTRODUCTION TO COMPUTERS

Overview

- Introductory definitions
- Components of a computer
- Parts of a computer
- Exploring the desktop
- Files and Folders
- Using the mouse
- Using the keyboard

Computer

- An electronic device that processes data according to a set of stored instructions.
- An electronic device that follow instructions to accept input (data), process the input and then produce output (information).
- Work through an interaction of hardware and software
  - **Hardware** - the physical parts of a computer that you can see and touch.
  - **Software** refers to programs that governs the operations of the computer.

Parts of a computer

- Hardware
  - Physical parts of a computer that you can see and touch
  - E.g. Monitor (screen), keyboard, mouse etc.
  - Internal hardware is found inside the main box of the computer
  - External hardware is found outside the system unit and plug into ports located on the exterior of the system unit.
- Software
  - Programs that control or maintain the operations of the computer and its devices.
  - Instructions or programs that perform actions on a hardware
  - Software is needed to control hardware and to turn it into a useful machine.

Software

- Two types:

1. Systems software
    - Programs that control or maintain the operations of the computer and its devices. Sub-divided into two:
    - **Operating Systems (OS):** is a set of programs that coordinates all activities among computer hardware devices and allows users to run application software e.g..
        - OS provides an interface that helps the user to interact with the computer.
        - OS boots the computer and launches programs at the user`s direction.
        - Examples of OS are Windows XP,7,8, 10,11, Linux, Macintosh, Android etc.
    - **Utility programs:** Programs that performs a very specific task related to managing system resources.
        - Allow the user to perform maintenance-type tasks related to managing a computer, its devices or its programs
        - Help to perform maintenance or correct problems with a computer system.
        - Examples: Virus scanner, backup software, data compression, system monitors, disk defragmenters etc.

Software cont…

2. Applications software
   - Programs that perform specific tasks for users. Enable users to solve problems, perform work or entertain themselves.

   Examples:
   - Word processor e.g. Microsoft Word
   - Spreadsheet e.g. Microsoft Excel
   - Presentation e.g. Microsoft PowerPoint
   - Viewing Web pages e.g. Mozilla Firefox, Chrome, Edge etc
   - Sending and receiving e-mail

Components of a computer

- An **input** (device) is a hardware used to enter data into the computer
- An **output** (device) is a hardware that conveys information to the user
- A **Central Processing Unit** (CPU) or **processor** changes data into information
- **Memory -** Holds the programs and data that users are working on.
- A **storage** stores data or information

Input devices

- Any device that enables users to enter data and instructions into the computer
  - Keyboard
    - Used to type information into your computer
  - Mouse
    - Allows data input by clicking to select options
  - Scanners
  - Barcode readers
  - Digital camera
  - Microphone

Input Devices

Hardware used to enter data and instructions



Output devices

- Any device that help users get information out of the computer in a meaningful way.
  - Monitor, screen or Visual Display Unit (VDU)
    - Cathode Ray Tube (CRT)
      - A vacuum tube used as a display screen in a computer or video display terminal.
      - Found in older desktop computers but also in older television sets.
    - Liquid Crystal Display (LCD)
      - A flat panel technology
  - Printers
  - Speakers
    - To deliver an audio output such as music, that is connected to the computer.

Output devices

Hardware that conveys information to a user



Storage devices

- Holds data, instructions, and information for future use
- Examples of storage devices are:
  - Hard disk drive
  - Compact disks
  - Memory card
  - Flash disks



Storage devices











Memory

- Temporary holding place for data and instructions when the processor (CPU) is using them.
- Where programs and data are kept when the processor is actively using them.
- A special chip that hold data and instructions ready for use by the CPU.
- **Types of Memory**
  - **RAM (Random Access Memory)**
  - **ROM (Read Only Memory)**



Random Access Memory (RAM)

- Also referred to as Main memory
- Holds the programs and data that users are working on.
- Temporarily hold (1) software instructions and (2) data before and after it is processed by the CPU.
- RAM is volatile – when the computer is switched off anything in RAM is lost.
- Data stored on disk is transferred into RAM when you open a file to work on it.

Read Only Memory (ROM)

- Memory that stores instructions and data that can be only be read, but it cannot be updated or modified.
- Important program instructions needed to start a computer and load OS are stored in ROM.
- ROM is **non-volatile** storage, which means that the information is maintained even if the component loses power.
- ROM plays a critical part in starting up the computer.
  - This is also referred to as **booting up** the computer.
- Modifying it is either impossible or very difficult.

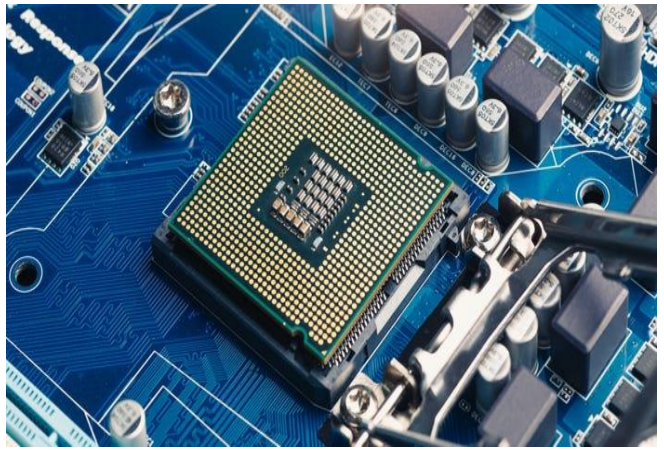
Central Processing Unit (CPU)

- Also called the processor or microprocessor.
- A part of a computer that controls all other parts.
- CPU interprets and carries out the basic instructions that operate a computer.
- Directs and coordinates most of the operations in a computer
- Carries out instructions that tell the computer what to do.
- Considered to be the brain of the computer.

Central Processing Unit (CPU)

- Also called the processor or microprocessor.





Types of computers

Desktop computers (personal Computer

- Designed for use at a desk or table
- Designed so that all of the components fit on or under a desk or table





Types of computers

Laptops and notebooks

- Lightweight mobile computers with a thin screen.
- Can operate on batteries, so you can take them anywhere.
- Unlike desktops, laptops combine CPU, screen, and keyboard in a single case
- Notebooks are small, affordable laptops that are designed to perform a limited number of tasks such as to browse the web and check e-mail.



Types of computers

Handheld computers (Palmtops)

- Also called personal digital assistants (PDAs)
- Are battery-powered computers small enough to carry almost anywhere.
- Useful for scheduling appointments, storing addresses and phone numbers, and playing games.
- Some have more advanced capabilities, such as making telephone calls or accessing the Internet.
- Have touch screens that you use with your finger or a stylus (a pen-shaped pointing tool).







Types of computers

Tablet PCs

- Mobile computers that combine features of laptops and handheld computers.
- Users write notes or draw pictures on the screen with a tablet pen instead of a stylus.
- Can also convert handwriting into typed text.
- Some Tablet PCs are “convertibles” with a screen that swivels and unfolds to reveal a keyboard underneath.





Types of computers

Smartphones

- Are mobile phones that have some of the same capabilities as a computer.
- Used to make telephone calls, access the Internet, organize contact information, send e-mail and text messages, play games and take pictures.
- Usually have a keyboard and a large screen.



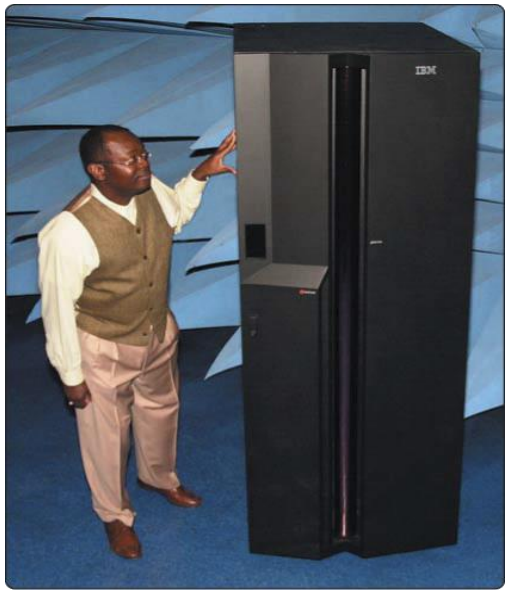
Types of computers

Server

- A computer that has been optimized to provide services to other computers over a network.
- Servers have powerful processors, lots of memory and large hard drives.
- Other types of computers can fill an entire room.
  - These are: ***Mainframes and Supercomputers***

Servers

Midrange server **Powerful, large computer that *supports up to a few thousand computers***



Mainframe **Very powerful, expensive computer that *supports thousands of computers. Used by large organizations***

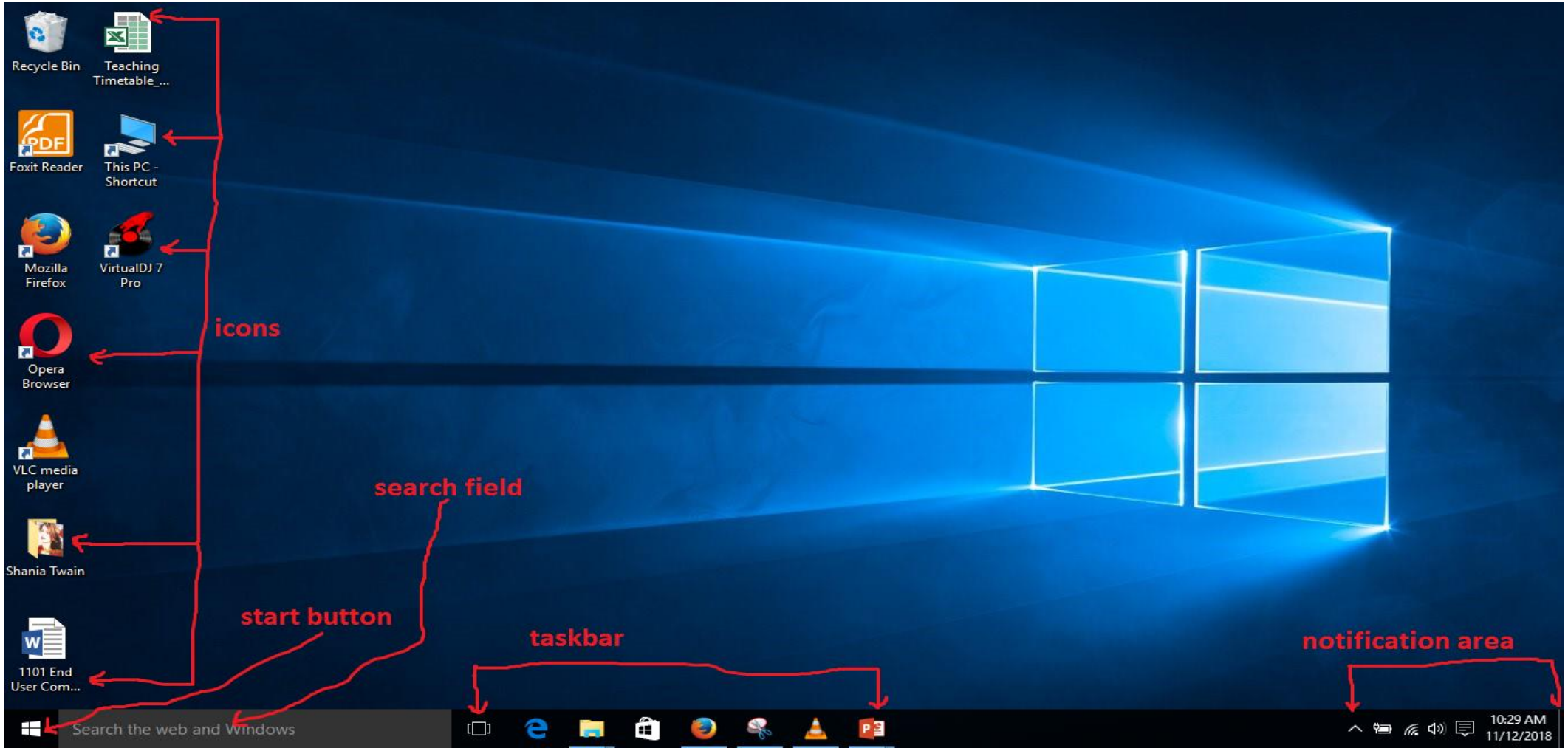
Supercomputer **The fastest, most powerful, most expensive computer. *Used for applications requiring complex mathematical calculations. Used in weather forecasting, climate research, oil and gas exploration, physical simulations etc.***



What can you do with computers?

- At workplace, people use computers to keep records, analyze data, do research, manage projects etc.
- At home, people use computers to find information, store pictures and music, track finances, play games and communicate with others.
- Send and receive messages by electronic mail (email
- Education – Online lessons
- Trade – Buying and selling of products and services.
- Connect to the Internet, send and receive emails etc.

Exploring the desktop in windows 10



The desktop

- The main screen area that you see after you turn on your computer and log on to Windows.
- It serves as a surface for your work.
- When you open programs or folders, they appear on the desktop.
- You can put things on the desktop, such as files and folders and arrange them the way you want.
- The taskbar sits at the bottom of your screen.
  - It shows you which programs are running and allows you to switch between them.
  - It also contains the Start button , which you can use to access programs, folders and computer settings.

The Taskbar

- Taskbar is the long horizontal bar at the bottom of your screen.
- Unlike the desktop, which can be covered by open windows,
- The taskbar is almost always visible. It has three main sections:
  - The **Start button**, which opens the Start menu.
  - The **middle section**, which shows you which programs and files you have open and allows you to quickly switch between them.
  - The **Notification area**, which includes a clock and icon (small pictures) that communicate the status of certain programs and computer settings.
- You're likely to use the middle section of the taskbar the most.

The Taskbar

- To switch to another window, click its Taskbar button.
- Clicking Taskbar buttons is one of several ways to switch between windows.
- When a window is active (its Taskbar button is highlighted),
- When you move mouse pointer to a Taskbar button, a small picture appears that shows a small or tiny version of the corresponding window.
- This preview, also called a **thumbnail,** is useful and if one of the windows has video or animation playing, you'll see it playing in the preview.

The Notification Area

- Notification area, on the far right side of the taskbar, includes a clock and a group of icons. It looks like this.



- These icons communicate the status of something on the computer or provide access to certain settings.
- The set of icons you see depends on which programs or services you have installed and how your computer manufacturer set up your computer.

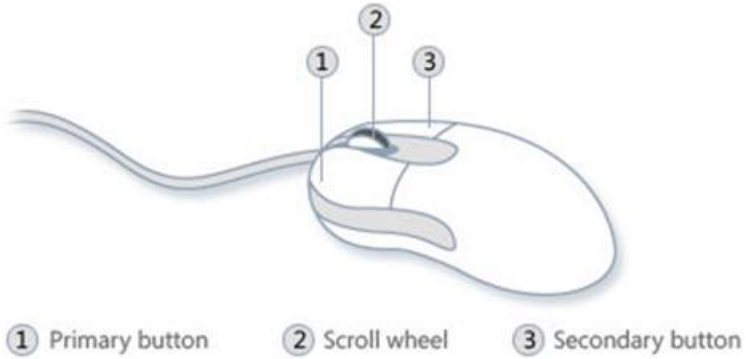
Using the mouse

- An input device used to enter data and instructions into the computer
- You can use the mouse to interact with items on the computer screen.
- You can move objects, open them, change them, throw them away, and perform other actions, all by pointing and clicking with the mouse.

Using the mouse

Basic parts

- A mouse has two buttons:
  - A primary button (usually the left button) and
  - A secondary button (usually the right button).
  - Primary button is mostly used.
  - Most mice also include a scroll wheel between the buttons to help users scroll through documents and webpages more easily.



Parts of a mouse

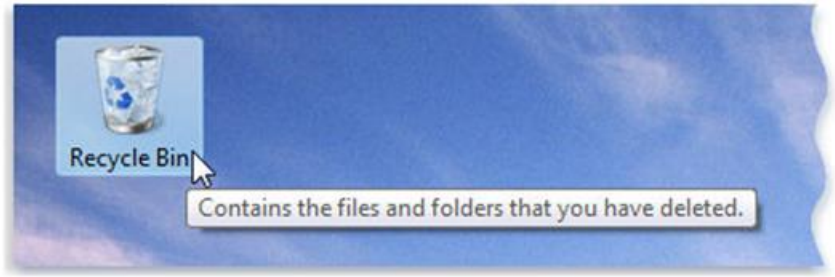
Holding and moving the mouse

- Place your mouse beside your keyboard on a clean, smooth surface, such as a mouse pad.
- Hold the mouse gently, with your index finger resting on the primary button and your thumb resting on the side.
- To move the mouse, slide it slowly in any direction.
- Don't twist it—keep the front of the mouse aimed away from you.
- As you move the mouse, a pointer (see picture) on your screen moves in the same direction.
- If you run out of room to move your mouse on your desk or mouse pad, just pick up the mouse and bring it back closer to you.

Pointing, clicking, and dragging

- Pointing to an item on the screen means moving your mouse so the pointer appears to be touching the item.
- When you point to something, a small box often appears that describes the item.
  - For example, when you point to the Recycle Bin on the desktop, a box appears with this information: "Contains the files and folders that you have deleted."

Holding and moving the mouse



Pointing to an object often reveals a descriptive message about it

- Mouse pointer can change depending on what you're pointing to. E.g., when you point to a link in web browser, the pointer changes from an arrow to a hand with a pointing finger .
- Most mouse actions combine pointing with pressing one of the mouse buttons.
- There are four basic ways to use your mouse buttons: clicking, double-clicking, right-clicking, and dragging.

Clicking and Double-clicking

Clicking (single-clicking)

- To click an item, point to the item on the screen, and then press and release the primary button (usually the left button).
- Clicking is most often used to select (mark) an item or open a menu. This is sometimes called single-clicking or left-clicking.

Double-clicking

- To double-click an item, point to the item on the screen, and then click twice quickly. If the two clicks are spaced too far apart, they might be interpreted as two individual clicks rather than as one double-click.
- Double-clicking is most often used to open items on your desktop. For example, you can start a program or open a folder by double-clicking its icon on the desktop.

Right-clicking

- To right-click an item, point to the item on the screen, and then press and release the secondary button (usually the right button).
- Right-clicking an item usually displays a list of things you can do with the item.
  - For example, when you right-click the Recycle Bin on your desktop, Windows displays a menu allowing you to open it, empty it, delete it, or see its properties. If you're ever unsure of what to do with something, right-click it.

Dragging

- You can move items around your screen by dragging them.
- To drag an object, point to the object on the screen, press and hold the primary button, move the object to a new location, and then release the primary button.
- Dragging (sometimes called dragging and dropping) is most often used to move files and folders to a different location and to move windows and icons around on your screen.

Using the scroll wheel

- If your mouse has a scroll wheel, you can use it to scroll through documents and webpages. To scroll down, roll the wheel backward (toward you). To scroll up, roll the wheel forward (away from you).

Tips for using your mouse safely

- Holding and moving your mouse properly can help you avoid soreness or injury to your wrists, hands, and arms, particularly if you use your computer for long periods of time. Here are some tips to help you avoid problems:
  - Place your mouse at elbow level. Your upper arms should fall relaxed at your sides.
  - Don't squeeze or grip your mouse tightly. Hold it lightly.
  - Move the mouse by pivoting your arm at your elbow. Avoid bending your wrist up, down, or to the sides.
  - Use a light touch when clicking a mouse button.
  - Keep your fingers relaxed. Don't allow them to hover above the buttons.
  - When you don't need to use the mouse, don't hold it.
  - Take short breaks from computer use every 15 to 20 minutes.

The Keyboard

- Keyboard is the main way to enter information into the computer.
- You can also use the keyboard to control the computer.
- Learning just a few simple keyboard commands (instructions to the computer) can help you work more efficiently.

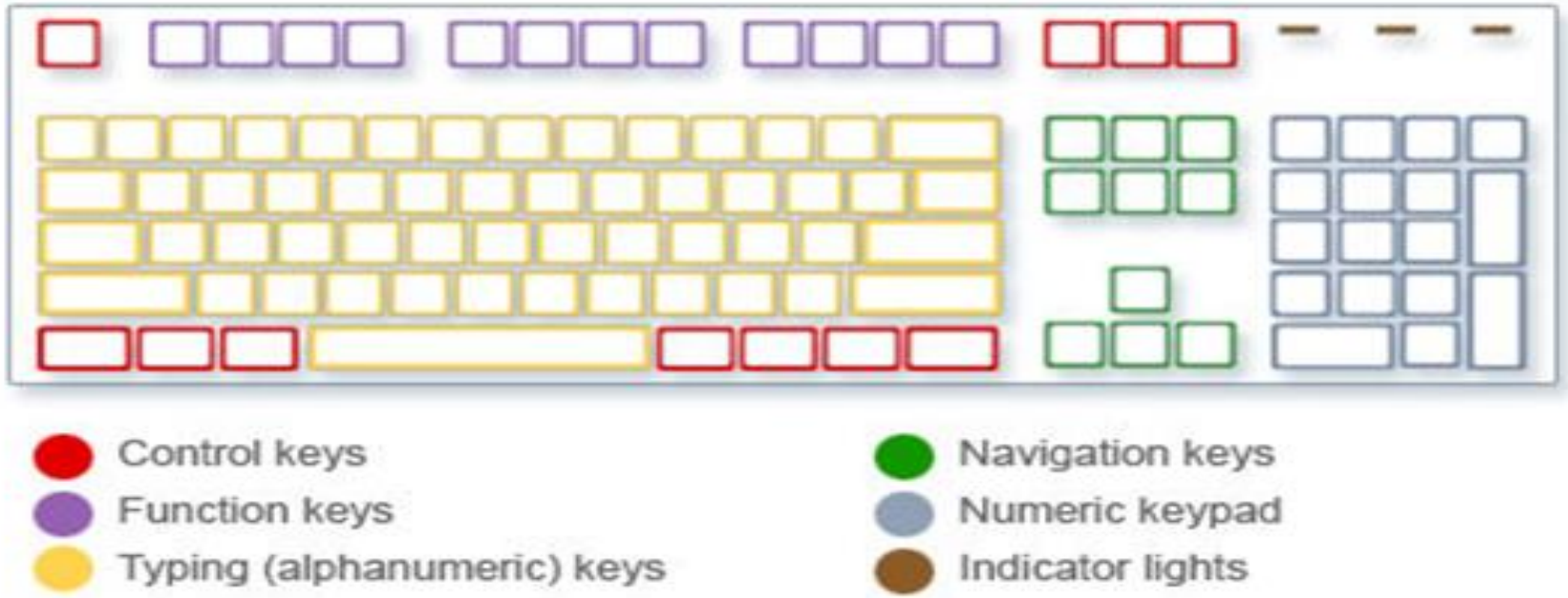
Keyboard

- Keyboard is the main way to enter information into the computer.
- You can also use the keyboard to control the computer.



Keyboard keys

- The following illustration shows how these keys are arranged on a typical keyboard. Some keyboard layout might be different.



How the keys are arranged on a keyboard

Keyboard keys

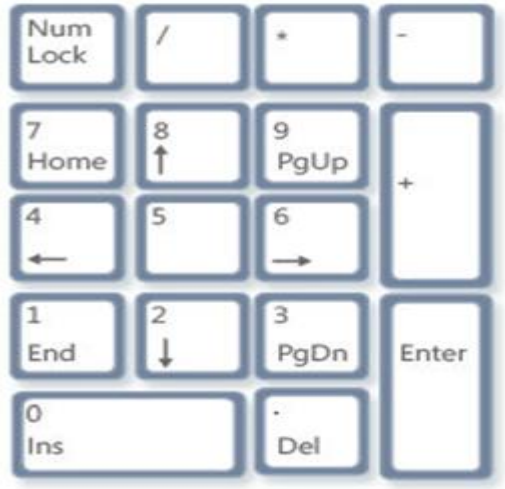
- The keys on the keyboard can be divided into several groups based on function:
- **Typing (alphanumeric) keys**. These keys include the same letter, number, punctuation, and symbol keys.
- **Control keys** - These keys are used alone or in combination with other keys to perform certain actions. The most frequently used control keys are Ctrl, Alt, the Windows logo key and Esc.
- **Function keys**. The function keys are used to perform specific tasks. They are labeled as F1, F2, F3 and so on, up to F12. The functionality of these keys differs from program to program.
- **Navigation keys** - These keys are used for moving around in documents or webpages and editing text. They include the arrow keys, Home, End, Page Up, Page Down, Delete, and Insert.
- **Numeric keypad** - The numeric keypad is handy for entering numbers quickly. The keys are grouped together in a block like a conventional calculator or adding machine.

Press this	To do this
Left Arrow, Right Arrow, Up Arrow, or Down Arrow	Move the cursor or selection one space or line in the direction of the arrow, or scroll a webpage in the direction of the arrow
Home	Move the cursor to the beginning of a line or move to the top of a webpage
End	Move the cursor to the end of a line or move to the bottom of a webpage
Ctrl+Home	Move to the top of a document
Ctrl+End	Move to the bottom of a document
Page Up	Move the cursor or page up one screen
Page Down	Move the cursor or page down one screen
Delete	Delete the character after the cursor, or the selected text; in Windows, delete the selected item and move it to the Recycle Bin
Insert	Turn Insert mode off or on. When Insert mode is on, text that you type is inserted at the cursor. When Insert mode is off, text that you type replaces existing characters.

Key name	How to use it
Shift	Press Shift in combination with a letter to type an uppercase letter. Press Shift in combination with another key to type the symbol shown on the upper part of that key.
Caps Lock	Press Caps Lock once to type all letters as uppercase. Press Caps Lock again to turn this function off. Your keyboard might have a light indicating whether Caps Lock is on.
Tab	Press Tab to move the cursor several spaces forward. You can also press Tab to move to the next text box on a form.
Enter	Press Enter to move the cursor to the beginning of the next line. In a dialog box, press Enter to select the highlighted button.
Spacebar	Press the Spacebar to move the cursor one space forward.
Backspace	Press Backspace to delete the character before the cursor, or the selected text.

Using the numeric keypad

- The numeric keypad arranges the numerals 0 though 9, the arithmetic operators + (addition), - (subtraction), * (multiplication), and / (division), and the decimal point as they would appear on a calculator or adding machine.
- These characters are duplicated elsewhere on the keyboard, but the keypad arrangement allows you to quickly enter numerical data or mathematical operations with one hand.



Numeric keypad

Using the numeric keypad

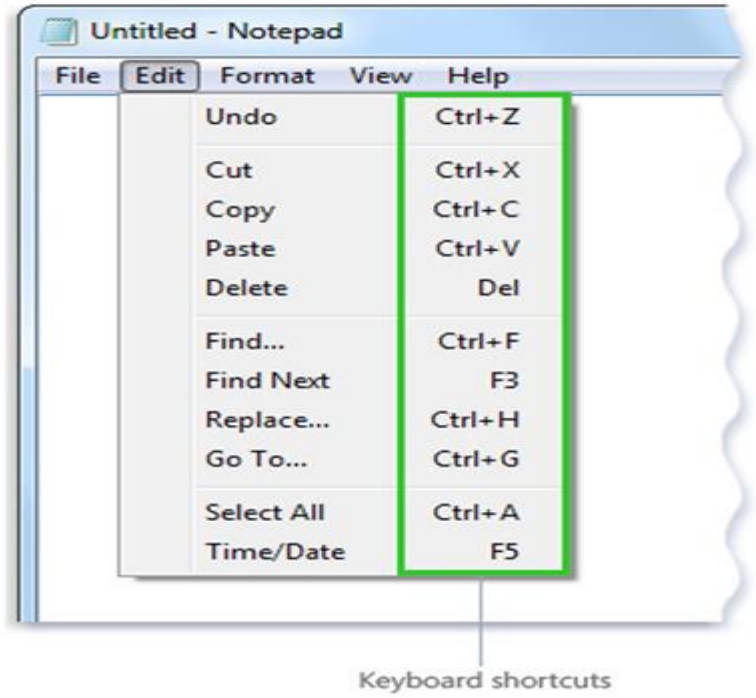
- To use the numeric keypad to enter numbers, press Num Lock.
- Most keyboards have a light that indicates whether Num Lock is on or off.
- When Num Lock is off, the numeric keypad functions as a second set of navigation keys (these functions are printed on the keys next to the numerals or symbols).
- You can use numeric keypad to perform simple calculations with Calculator.

Using keyboard shortcuts

- Keyboard shortcuts are ways to perform actions by using your keyboard.
- They're called shortcuts because they help you work faster.
- Almost any action or command performed with a mouse can be performed faster using one or more keys on the keyboard.
- A plus sign (+) between two or more keys indicates that those keys should be pressed in combination.
  - E.g. Ctrl+A means to press and hold Ctrl and then press A. Ctrl+Shift+A means to press and hold Ctrl and Shift and then press A.

Find program shortcuts

- You can do things in most programs by using the keyboard. To see which commands have keyboard shortcuts, open a menu. The shortcuts (if available) are shown next to the menu items.



Keyboard shortcuts appear next to menu items.

Tips for using your keyboard safely

- Using your keyboard properly can help avoid soreness or injury to your wrists, hands, and arms, particularly if you use your computer for long periods of time.
- Here are some tips to help you avoid problems:
- Place your keyboard at elbow level. Your upper arms should be relaxed at your sides.
- Center your keyboard in front of you. If your keyboard has a numeric keypad, you can use the spacebar as the centering point.
- Type with your hands and wrists floating above the keyboard, so that you can use your whole arm to reach for distant keys instead of stretching your fingers.
- Avoid resting your palms or wrists on any type of surface while typing. If your keyboard has a palm rest, use it only during breaks from typing.
- While typing, use a light touch and keep your wrists straight.
- When you're not typing, relax your arms and hands.
- Take short breaks from computer use every 15 to 20 minutes.

Typing text

- Whenever you need to type something in a program, e-mail message or text box, you'll see a **blinking vertical line (I).**
- That's the cursor also called the **insertion point.**
- It shows where the text that you type will begin.
- You can move the cursor by clicking in the desired location with the mouse, or by using the navigation keys.
- In addition to letters, numerals, punctuation marks, and symbols, Typing keys also include Shift, Caps Lock, Tab, Enter, the Spacebar and Backspace.

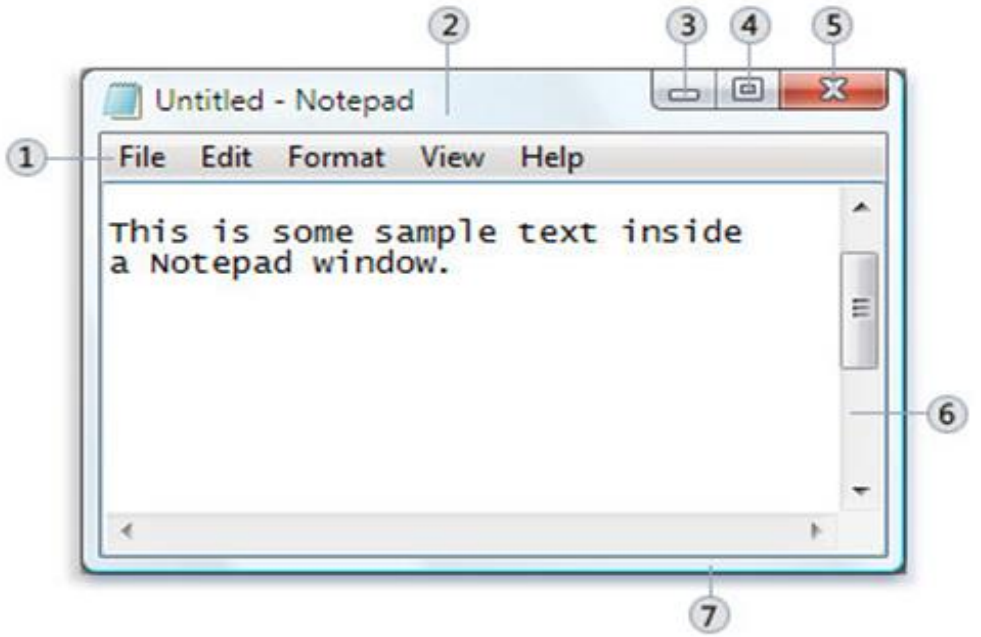
Working with windows

- When you open a program, file or folder, it appears on the screen in a box or frame called a window
  - (that's where the Windows operating system gets its name).
- A window is a rectangle on your screen that organizes how you view and work in specific programs or files on your computer.

- See the picture in the next slide.

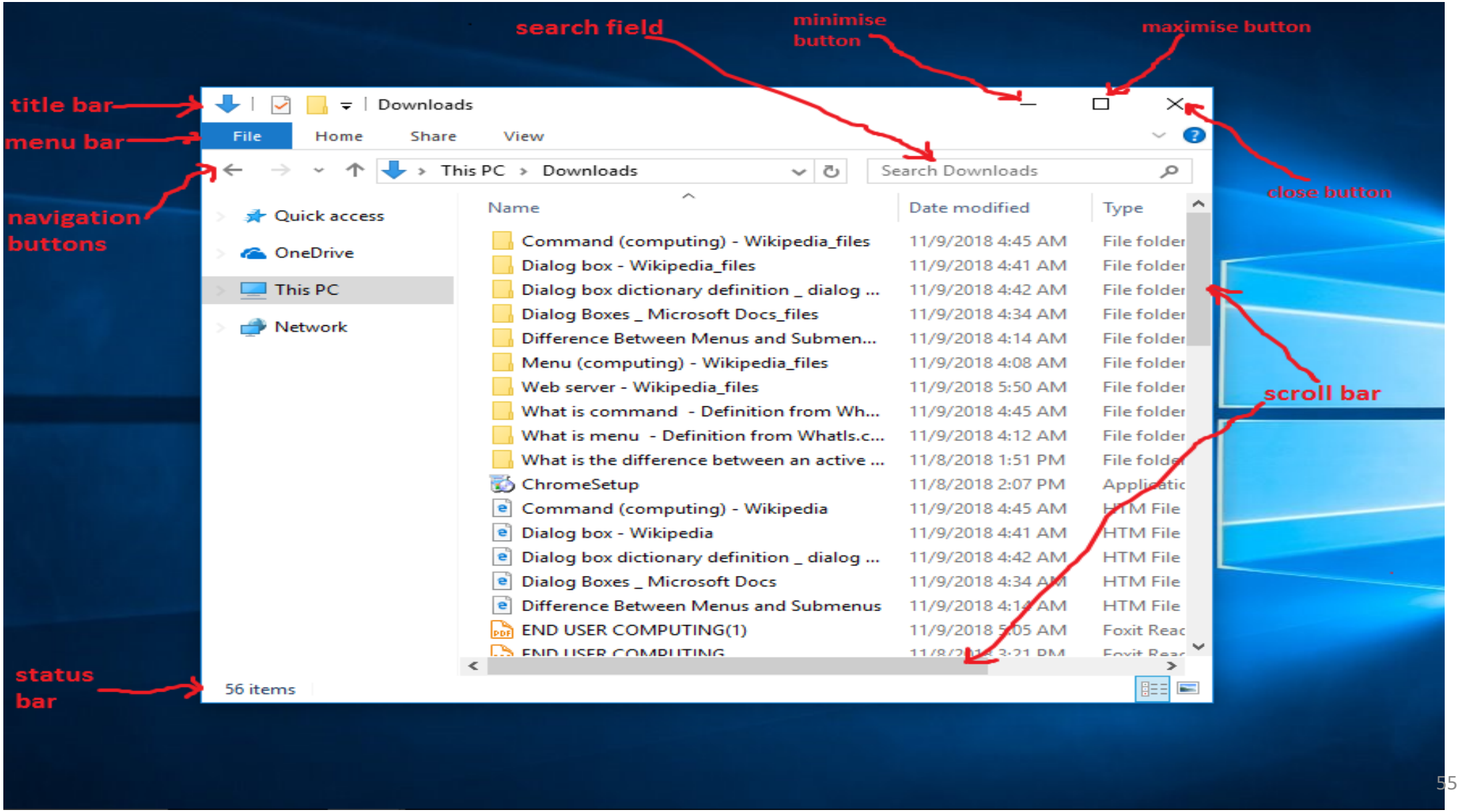
Working with windows

- All windows share some things in common.
- Windows always appear on the *desktop*—the main work area of your screen.
- Most windows have the same basic parts.



Parts of a typical window

Working with windows

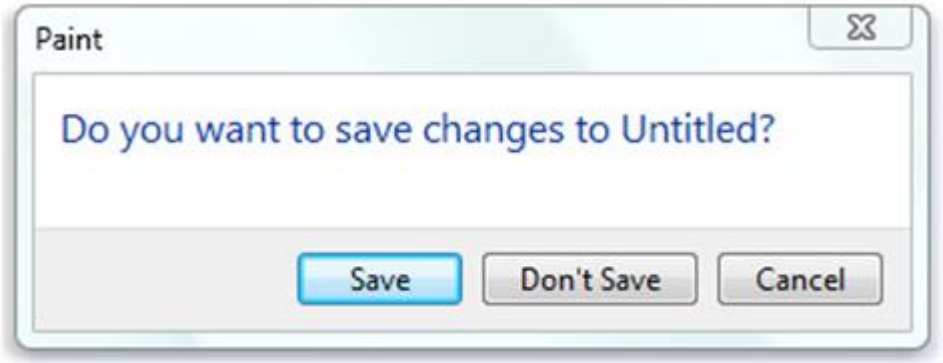


Parts of a window

- **Title bar** - Displays the name of the document and program (or the folder name if you're working in a folder).
- **Minimize, Maximize, and Close buttons**. These buttons hide the window, enlarge it to fill the whole screen, and close it, respectively
- **Menu bar** - Contains items that the can click to make choices in a program.
- **Scroll bar** – Help users to scroll the contents of the window to see information that is currently out of view.
- **Borders and corners** - You can drag these with your mouse pointer to change the size of the window.
- Other windows may have additional buttons, boxes, or bars. But they'll usually have the basic parts, too.

Dialog Boxes

- A special type of window that ask users a question,
- Allows users to select options to perform a task, or provides users with information.
- You'll see dialog boxes when a program or Windows needs a response from the user before it can continue.

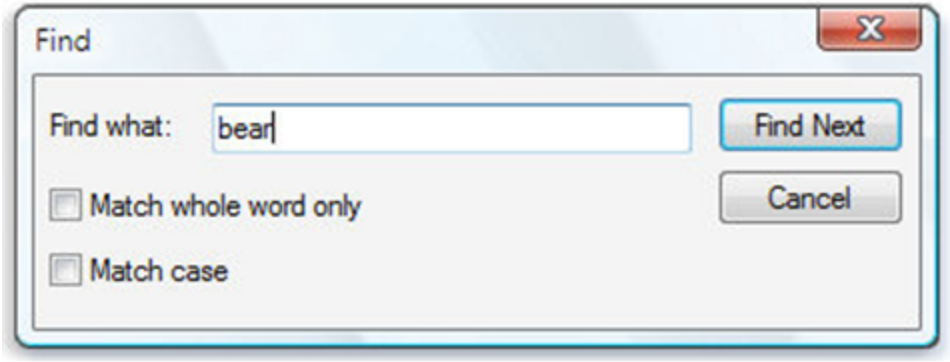


A dialog box appears if you exit a program without saving your work

- Most dialog boxes can't be maximized, minimized, or resized. They can, however, be moved.

Using Text Boxes

- A text box allow users to type information, such as a search term or password.
- The following picture shows a dialog box containing a text box. We've entered "bear" into the text box.

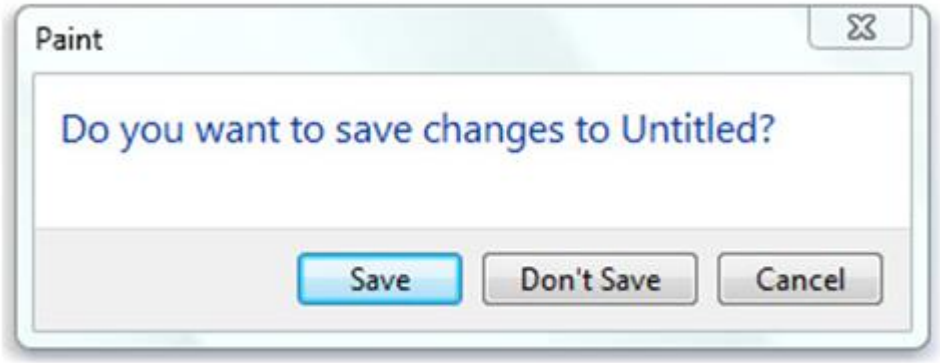


Example of a text box in a dialog box

- A blinking vertical line called the cursor indicates where text that you type will appear. In the example, you can see the cursor after the "r" in "bear." You can easily move the cursor by clicking the new position. For example, to add a word before "bear," you would first move the cursor by clicking before the "b."
- If you don't see a cursor in the text box, it means the text box isn't ready for your input. Click the box first, and then start typing.

Using Command Buttons

- A command button performs a command (makes something happen) when you click it.
- You'll see them in dialog boxes, which are small windows that contain options for completing a task.
- E.g. if you close a Paint picture without saving it first, you might see a dialog box like this.



Dialog box with three buttons

Working with files and folders

- A **file** is a document or collection of data that has a name and is stored in a computer
  - E.g. music, text, image, and video files
- A **folder** is a directory used to store/organize your files
- Files and folders help to keep your computer organized
- Computer documents are saved as files
- Folders are used to store/organize your files

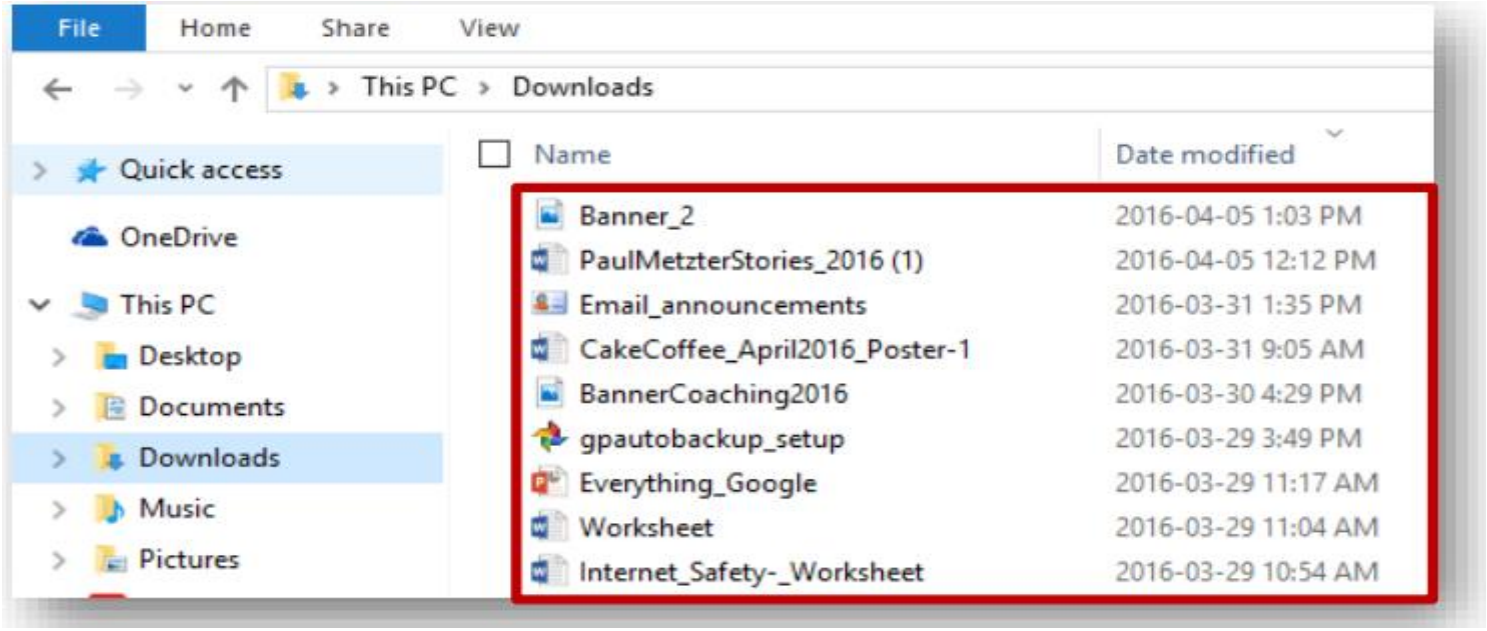
Working with files and folders

- A file is an item that contains information—for example, text, images or music.
- On computers, files are represented with icons; this makes it easy to recognize a type of file by looking at its icon.
- The common way to create new files is by using a program.
  - E.g. you can create a text document in a word-processing program
- Here are some common file icons:



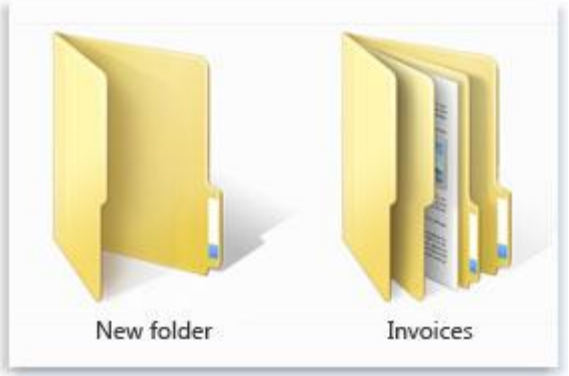
Files

- Computer documents are saved as files (text, image, audio, video)
- Examples of some files are:
  - A picture you saved/downloaded, a letter, a song or video you downloaded



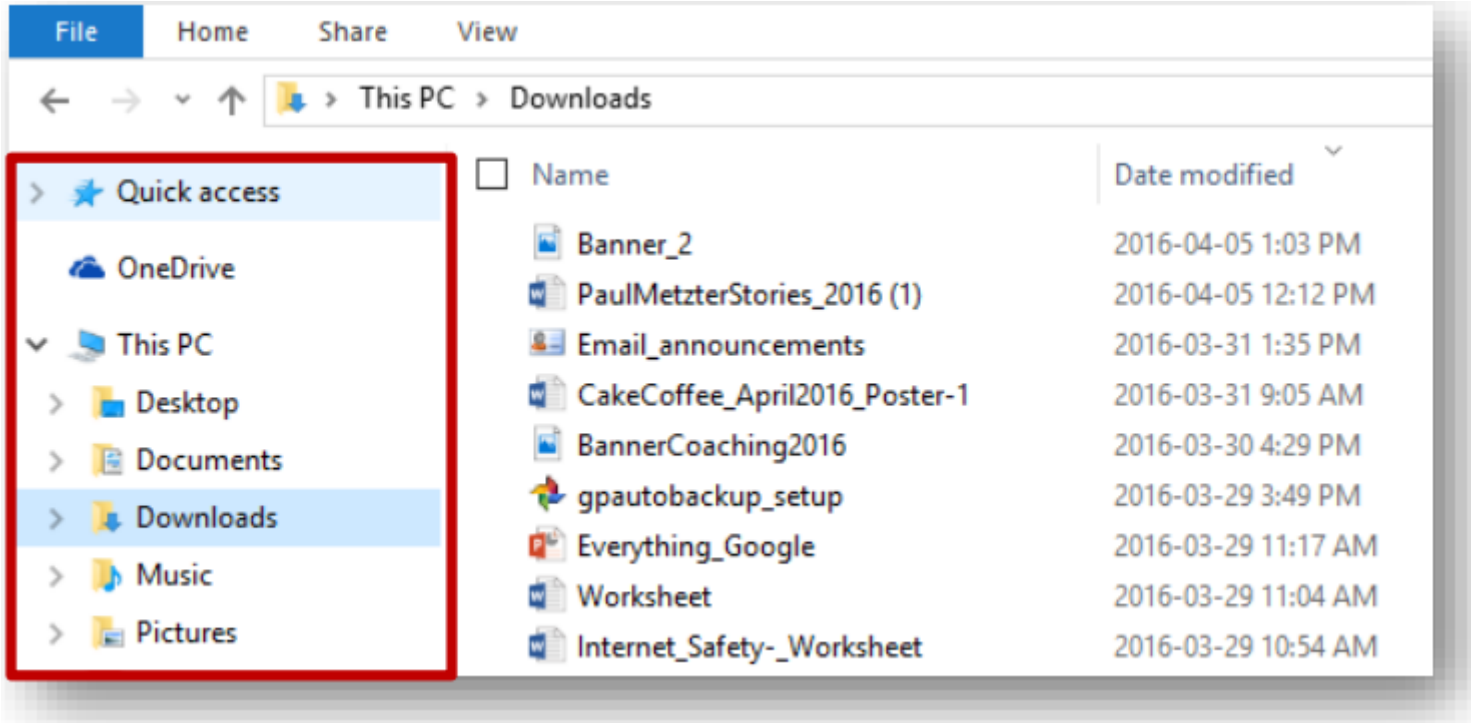
Folders

- A folder is a container you can use to store files in.
- Folders can also store other folders.
- A folder within a folder is usually called a subfolder.
- You can create any number of subfolders, and each can hold any number of files and additional subfolders.
- Here are some typical folder icons:



Folders

- Folders are used to store your files. They are like File Cabinets, containing drawers, which contain file folders that can contain many documents.



Libraries

- Windows Operating System has some folders already set up for you in your PC:
  - Documents
  - Music
  - Pictures
  - Videos
- You can organize your own documents into these folders

Using libraries to access your files and folders

- You can use libraries to access your files and folders, and arrange them in different ways.
- Four default libraries and what they're used for:
  - Documents library.
    - Use to organize and arrange word-processing documents, spreadsheets, presentations, and other text-related files.
  - Pictures library.
    - Use to organize and arrange your digital pictures, whether you get them from your camera, scanner, or in e-mail from other people..
  - Music library.
    - Use to organize and arrange your digital music, such as songs that you rip from an audio CD or that you download from the Internet.
  - Videos library.
    - Use to organize and arrange your videos, such as clips from your digital camera or video files that you download from the Internet.